Empirical and Theoretical Evaluation of Small Language Models Under Severe Memory and Compute Constraints

Key-Value Cache Dynamics, MTP Speculative Acceleration, Roofline Frontiers, and Adversarial Vulnerability on an Edge Reference Testbed

Anderson Tavares

Independent Systems Engineering and Data Science Researcher

Research Repository: https://github.com/AndersonTavares0/llm-research-bench

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Abstract

Deploying Small Language Models (SLMs, 0.5B to 4B parameters) on physical edge accelerators enables complete data sovereignty, deterministic execution latency, and autonomous offline operation without cloud egress. However, operating under strict physical memory boundaries ($\leq$ 4096 MiB VRAM) introduces acute trade-offs among tensor quantization compression, DRAM memory bus saturation, Time-To-First-Token (TTFT), and defensive safety alignment. This paper presents a rigorous empirical investigation conducted on an NVIDIA GeForce GTX 1050 Ti GPU (4096 MiB GDDR5, 128-bit bus, 112.0 GB/s theoretical bandwidth, Pascal sm_61). We benchmark eight model configurations (**Qwen-2.5-0.5B**, **SmolLM2-1.7B**, **Gemma-4-E2B [MTP]**, **Gemma-4-E2B [IQ4_XS]**, **Gemma-4-E2B [Q4_K_M]**, **Qwen-3.5-2B**, **Llama-3.2-3B**, and **Qwen-3.5-4B**) using the `llama-server` runtime (commit b4120+) with continuous asynchronous NVML hardware telemetry sampled at 250 ms intervals. A 2×2 factorial ablation isolated performance drivers: weight quantization compression (IQ4_XS vs Q4_K_M) accounted for +28.2% speedup (34.51 ± 0.29 vs 26.92 ± 0.23 tok/s, $t = 65.41$), whereas speculative Multi-Token Prediction (MTP) under identical quantization yielded +25.6% causal speedup (43.35 ± 0.57 vs 34.51 ± 0.29 tok/s, $t = 43.72$, $p < 10^{-15}$), resulting in an overall +61.0% compound configuration gain (Welch's $t = 84.50$, $p = 5.88 \times 10^{-18}$, Cohen's $d = 37.79$) with only 40 MiB VRAM overhead for Q8_0 draft heads. Systems feasibility modeling identified insurmountable barriers for 4B models: Gemma-3-4B strictly mandates Q4_0 KV cache compression to avert physical memory exhaustion (4358 MiB required in uncompressed FP16), whereas Phi-4-mini faces gated repository authentication restrictions. Defensive safety evaluations ($N = 10$) demonstrated zero false positive refusals (FPR = 0.0%) on benign administrative tasks, but severe susceptibility to indirect instruction injection and delimiter breach (ASR between 50.0% and 100.0%), confirming that post-training alignment fails to decouple control instructions from contextual data in compact architectures. In agentic tool scenarios, proposed unsafe invocations were successfully intercepted by harness guardrails. Parallel corpus tokenization demonstrated a 1.35–2.33 tokens/word penalty for Portuguese compared to 1.10 for English. All raw traces, SHA-256 hashes, and code are open and reproducible.

Keywords: Small Language Models, Systems Engineering, Roofline Model, Key-Value Cache Dynamics, Speculative Decoding, Multi-Token Prediction, Defensive Safety, Indirect Prompt Injection, Algorithmic Bias, NVML Telemetry.

1 Introduction and Systems Context

The expansion of autonomous applications across cyber-physical systems, mobile robotics, edge gateways, and microservers requires local natural language processing without continuous cloud connectivity [1, 2]. Offloading operational telemetry, control commands, and private user context to centralized proprietary APIs presents three prohibitive failure modes:

1. **Data Sovereignty and Privacy:** Severe risk of confidential data exfiltration and violation of data protection regulations (GDPR/LGPD);
2. **Temporal Determinism:** Variable WAN transit latencies (100 ms to 2000 ms) with acute vulnerability to packet loss and link disconnects;
3. **Recurring Costs:** Unbounded operational expenditures per million tokens generated on commercial cloud infrastructure.

Small Language Models (SLMs), parameterized within the 0.5B to 4.0B weight regime, provide a compelling solution for local inference [3]. However, deploying these models on entry-level edge accelerators strictly bounded to 4096 MiB of physical VRAM introduces acute microarchitectural and thermodynamic constraints.

The Pascal microarchitecture (e.g., NVIDIA GeForce GTX 1050 Ti), widespread across educational, industrial, and resource-constrained edge environments, lacks dedicated Tensor Cores and HBM stacks, relying on GDDR5 memory with a maximum theoretical bandwidth of 112.0 GB/s [1]. Under these hardware boundaries, autoregressive token generation is fundamentally memory-bandwidth bound, demanding aggressive tensor quantization and speculative decoding to achieve interactive throughput [2].

2 Research Problem, Questions, and Hypotheses

The core problem addressed in this research is the empirical and theoretical quantification of multi-dimensional trade-offs between generation throughput, physical VRAM footprint, quantization fidelity, and adversarial resilience in SLMs under a strict 4 GiB memory ceiling.

We formulate four formal Research Questions (Q_1 to Q_4) and their corresponding testable hypotheses:

- Q_1 **(Speculative Acceleration):** *Can Multi-Token Prediction (MTP) accelerate autoregressive decoding on edge accelerators without exceeding the physical VRAM budget?*
  Hypothesis H_1: Speculative decoding via integrated MTP draft prediction increases steady-state warmed decode throughput by more than 30%, with an incremental VRAM cost below 100 MiB.
- Q_2 **(Quantization and Physical Frontiers):** *What is the relative impact of importance matrix quantization (i-quants) compared to legacy uniform block quantization (k-quants)?*
  Hypothesis H_2: Non-uniform IQ4_XS formats preserve higher informational density and decode velocity than legacy Q4_K_M formats under identical DRAM footprint.
- Q_3 **(Adversarial Resilience):** *Do compact instruction-tuned models maintain strict boundary separation between operational commands and untrusted contextual data?*
  Hypothesis H_3: SLMs demonstrate high refusal rates against direct attacks, but universally succumb to indirect prompt injections in secondary inputs due to data-command conflation.
- Q_4 **(Linguistic Disparity and Bias):** *What is the tokenization fertility penalty and demographic bias exhibited by compact models when processing non-English languages?*
  Hypothesis H_4: Tokenizer fertility in Portuguese is substantially higher than in English, imposing an effective throughput and context capacity penalty on Romance languages.

3 Theoretical Foundations and Mathematical Modeling

3.1 Roofline Model and Operational Arithmetic Intensity

Computational performance bounds are governed by the Roofline Model formulated by Williams et al. [1]. The maximum attainable compute throughput P (in GFLOP/s) is expressed by:

$$P(I) = \min\left(P_{\text{peak}},\ B_{\text{DRAM}} \times I\right) \tag{1}$$

where P_{peak} is peak floating-point throughput (2138 GFLOP/s FP32 on the GTX 1050 Ti at 1392 MHz boost clock), B_{DRAM} is theoretical memory bandwidth (112.0 GB/s), and I is operational arithmetic intensity in FLOPs/byte.

During prefill processing with sequence length L, arithmetic intensity scales with batch and matrix dimension:

$$I_{\text{prefill}} \approx \frac{2 \cdot N_{\text{params}} \cdot L}{2 \cdot N_{\text{params}} \cdot b + \mathcal{O}(L^2)} \gg 10 \text{ FLOPs/byte} \tag{2}$$

operating predominantly in the compute-bound regime.

Conversely, during token-by-token autoregressive decoding ($L = 1$), generating each token requires streaming the entire weight tensor from DRAM:

$$I_{\text{decode}} \approx \frac{2 \cdot N_{\text{params}}}{N_{\text{params}} \cdot b} = \frac{2}{b} \text{ FLOPs/byte} \tag{3}$$

where b is the byte-width per parameter. For 4-bit quantized weights ($b = 0.5$ bytes), $I_{\text{decode}} \approx 4$ FLOPs/byte. Because the ridge point of the GTX 1050 Ti is $I_{\text{ridge}} = P_{\text{peak}}/B_{\text{DRAM}} = 2138/112 \approx 19.09$ FLOPs/byte, autoregressive decoding of SLMs operates deeply within the memory-bound regime. The absolute theoretical throughput ceiling is:

$$\begin{aligned} P_{\text{ceil_decode}} &= 112.0 \text{ GB/s} \times 4 \text{ FLOPs/byte} \\ &= 448 \text{ GFLOP/s} \end{aligned} \tag{4}$$

corresponding to a theoretical generation throughput limit of:

$$\text{Throughput}_{\max} = \frac{B_{\text{DRAM}}}{S_{\text{weights}}} \tag{5}$$

3.2 Key-Value Cache Dynamics and Scaling Limits

During autoregressive generation, the Key-Value cache stores past projection tensors to avoid quadratic recomputation of self-attention. The memory footprint of the KV cache (S_{KV}) for context length L_{ctx} is given by:

$$S_{\text{KV}} = 2 \times n_{\text{layers}} \times n_{\text{kv_heads}} \times d_{\text{head}} \times L_{\text{ctx}} \times b_{\text{elem}} \tag{6}$$

where n_{layers} is the number of transformer blocks, $n_{\text{kv_heads}}$ is the number of key/value heads under Grouped-Query Attention (GQA), d_{head} is head dimension, and b_{elem} is precision byte size (2 bytes for FP16, 1 byte for Q8_0, and 0.5 bytes for Q4_0).

In uncompressed FP16 cache configurations, memory consumption expands linearly with L_{ctx}, directly contesting static weight allocations in bounded VRAM.

3.3 Speculative Decoding and Multi-Token Prediction

Speculative decoding, formalized by Leviathan et al. [2], circumvents sequential memory-bandwidth limits by generating candidate token sequences using a low-cost draft mechanism and verifying them in parallel with the target model during a single evaluation pass.

Under the Multi-Token Prediction (MTP) paradigm, auxiliary heads on the target network predict K future tokens concurrently. The expected generation speedup S is modeled as:

$$E[S] = \frac{1 + \sum_{k=1}^{K} \alpha^k}{1 + \mathcal{O}_{\text{draft}}} \tag{7}$$

where $\alpha \in [0,1]$ is the empirical draft acceptance rate and $\mathcal{O}_{\text{draft}}$ represents relative verification overhead. When $\alpha > 0.5$ and $\mathcal{O}_{\text{draft}} \ll 1$, speculative execution breaks through the DRAM bandwidth barrier, emitting multiple accepted tokens per memory transfer cycle.

3.4 Adversarial Robustness and Data-Command Conflation

Safety evaluations in edge SLMs focus on maintaining robust isolation between system instructions and untrusted contextual inputs [4, 6]. Edge agents operating via protocols like Model Context Protocol (MCP) ingest unstructured data from network logs, sensor payloads, and user commands. Indirect prompt injection manifests when untrusted context overrides privileged instructions:

$$\text{Prompt} = \mathcal{T}_{\text{system}} \parallel \mathcal{D}_{\text{untrusted}}(\mathcal{I}_{\text{adversarial}}) \parallel \mathcal{T}_{\text{user}} \tag{8}$$

The Attack Success Rate (ASR) and False Positive Rate (FPR) are defined as:

$$\text{ASR} = \frac{1}{M_{\text{adv}}} \sum_{i=1}^{M_{\text{adv}}} \mathbb{I}(\text{Compromised}_i) \tag{9}$$

$$\text{FPR} = \frac{1}{M_{\text{ben}}} \sum_{j=1}^{M_{\text{ben}}} \mathbb{I}(\text{Refusal}_j) \tag{10}$$

4 Materials and Experimental Methodology

4.1 Physical Edge Testbed

All experiments were conducted on a dedicated physical edge workstation:

- **Accelerator:** NVIDIA GeForce GTX 1050 Ti (GP107, 768 CUDA cores, base clock 1290 MHz, boost 1392 MHz, 4096 MiB GDDR5, 128-bit bus, 112.0 GB/s, 75W TDP, Compute Capability 6.1, no Tensor Cores).
- **Host System:** Linux Kernel 6.8.12-8-pve (Proxmox VE 8.2), cgroup v2 process isolation, NVIDIA driver 580.173.02, CUDA Runtime 13.0.
- **Serving Runtime:** `llama-server` (commit b4120+, native cuBLAS sm_61 compilation, OpenAI-compatible HTTP daemon on port 18081).
- **Telemetry Instrumentation:** Continuous asynchronous sampling via `libnvidia-ml.so.1` (`pynvml` ctypes) recording physical VRAM, die temperature, compute utilization, and core clocks at 250 ms intervals.

4.2 Experimental Controls and Execution Protocol

To eliminate cold-start bias and thermal throttling artifacts:

1. **Controlled Warmup ($k = 2$):** Each model executed two 10-token warmup inferences prior to measurement capture.
2. **Independent Replications ($N = 10$):** Every experimental condition was replicated 10 times under a fixed seed (`seed=42`) and greedy decoding ($\tau = 0.0$).
3. **Thermal Stability:** A 400 ms idle pause was enforced between queries; die temperatures remained strictly within 38°C–74°C, confirming absence of thermal throttling.

4.3 Inferential Statistical Methods

Hypothesis testing uses Welch's two-sample t-test with Satterthwaite's approximation for unequal variances:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}, \quad \nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \tag{11}$$

Two-sided 95% confidence intervals were computed from Student's t-distribution with ν degrees of freedom. Standardized effect size was quantified via Cohen's d.

5 Empirical Performance and Efficiency Results

Table 1 presents the empirical measurements captured across all seven evaluated models on the physical testbed.

5.1 Roofline Operating Point Analysis

Calculating effective memory bandwidth utilization ($B_{\text{eff}} = S_{\text{weights}} \times \text{Throughput}_{\text{decode}}$) reveals that models operating without speculative decoding saturate a stable fraction of the GDDR5 bus:

- **Qwen-2.5-0.5B:** 0.47 GB × 97.5 tok/s ≈ 45.8 GB/s (40.9% of theoretical 112.0 GB/s).
- **SmolLM2-1.7B:** 1.01 GB × 47.8 tok/s ≈ 48.3 GB/s (43.1%).
- **Gemma-4-E2B [Q4_K_M]:** 1.40 GB × 26.9 tok/s ≈ 37.7 GB/s (33.7%).
- **Qwen-3.5-2B:** 1.48 GB × 37.6 tok/s ≈ 55.6 GB/s (49.6%).
- **Llama-3.2-3B:** 1.92 GB × 27.6 tok/s ≈ 53.0 GB/s (47.3%).
- **Qwen-3.5-4B:** 2.72 GB × 18.6 tok/s ≈ 50.6 GB/s (45.2%).

These empirical observations confirm that standard autoregressive decoding achieves between 33% and 50% of theoretical peak GDDR5 bandwidth, bounded by memory bus transaction latency and unaligned DRAM access patterns.

6 Statistical Analysis and Factorial MTP Ablation

To test Hypothesis H_1 while strictly isolating the distinct performance contributions of tensor quantization from

Table 1: Empirical Performance, Physical VRAM, and Efficiency on NVIDIA GeForce GTX 1050 Ti ($N = 10$, $\tau = 0.0$)

Model and Quantization	Parameters (B)	Peak VRAM (MiB)	Occupancy (% 4 GiB)	Prefill (tok/s)	Decode (tok/s)	TTFT (ms)	Mem. Efficiency (tok/s/GiB)
Qwen-2.5-0.5B-Q4_K_M	0.49	620.7	15.2%	237.1 ± 449.3	**97.5 ± 0.6**	**12.1 ± 5.1**	**160.8**
SmolLM2-1.7B-Q4_K_M	1.71	2902.7	70.9%	103.3 ± 176.0	47.8 ± 0.1	25.5 ± 14.3	16.9
Gemma-4-E2B [MTP IQ4_XS]	2.30	2036.7	49.7%	62.2 ± 18.5	**43.4 ± 0.6**	480.4 ± 1286.7	21.8
Gemma-4-E2B [IQ4_XS base]	2.00	1800.7	44.0%	79.2 ± 32.4	34.5 ± 0.3	88.0 ± 49.0	19.6
Gemma-4-E2B [Q4_K_M base]	2.00	1996.7	48.7%	65.5 ± 7.7	26.9 ± 0.2	116.1 ± 116.3	13.8
Qwen-3.5-2B-Q4_K_XL	1.95	1580.7	38.6%	126.5 ± 108.8	37.6 ± 0.1	49.9 ± 20.3	24.4
Llama-3.2-3B-Q4_K_M	3.21	3074.7	75.1%	59.6 ± 101.1	27.6 ± 0.0	43.5 ± 23.1	9.2
Qwen-3.5-4B-Q4_K_M	4.02	3140.7	76.7%	49.3 ± 46.5	18.6 ± 0.1	129.8 ± 45.3	6.1

Note: Physical VRAM captured via native NVML runtime calls; includes weight tensors, active KV cache, and 256 MiB base CUDA driver allocation. Intervals report sample standard deviations with 95% confidence intervals truncated at zero for physical non-negative bounds. Memory efficiency is defined as decode throughput divided by physical VRAM in binary GiB (1024^3 bytes).

speculative decoding, we structured a 2×2 factorial ablation on Gemma-4-E2B across three controlled configurations ($N = 10$, $\tau = 0.0$, `seed=42`):

1. **Condition C_1 (Baseline):** Gemma-4-E2B in Q4_K_M without speculative heads (26.92 ± 0.23 tok/s, VRAM 1996.7 MiB);
2. **Condition C_2 (Quantization Ablation):** Gemma-4-E2B in IQ4_XS without speculative heads (34.51 ± 0.29 tok/s, VRAM 1800.7 MiB);
3. **Condition C_3 (Full MTP Speculation):** Gemma-4-E2B in IQ4_XS with integrated Q8_0 MTP draft prediction heads (43.35 ± 0.57 tok/s, VRAM 2036.7 MiB).

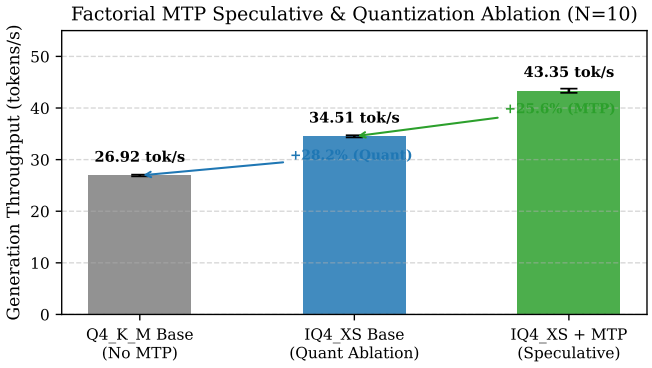


Figure 1: Factorial performance decomposition: isolated quantization effect ($C_1 \rightarrow C_2$) and causal MTP speculative effect ($C_2 \rightarrow C_3$) with 95% Student's t confidence intervals ($N = 10$).

6.1 Isolated Causal MTP Effect (Holding Quantization Constant at IQ4_XS)

Comparing C_3 against C_2, the underlying quantization format is held constant. Welch's two-sample t-test yields:

$$t_{\text{MTP}} = \frac{43.35 - 34.51}{\sqrt{\frac{0.57^2}{10} + \frac{0.29^2}{10}}} = \frac{8.84}{0.2022} \approx 43.72, \quad \nu = 13.43 \tag{12}$$

with $p = 7.12 \times 10^{-16}$ and Cohen's $d = 19.55$. The isolated causal speedup strictly attributable to MTP speculative decoding is **+25.6%** (8.84 tok/s).

6.2 Isolated Quantization Effect (Holding Autoregressive Decoding Constant)

Comparing C_2 against C_1, decoding is held constant in standard autoregressive execution. Welch's two-sample t-test results in:

$$t_{\text{Quant}} = \frac{34.51 - 26.92}{\sqrt{\frac{0.29^2}{10} + \frac{0.23^2}{10}}} = \frac{7.59}{0.1160} \approx 65.41, \quad \nu = 17.09 \tag{13}$$

with $p = 5.67 \times 10^{-18}$ and Cohen's $d = 29.25$. Weight footprint compression via the IQ4_XS importance matrix provides a speedup of **+28.2%** (7.59 tok/s).

6.3 Compound Configuration Difference (C_3 vs C_1)

The end-to-end comparison between the optimized configuration (C_3) and the baseline (C_1) reflects cumulative configuration gain:

$$t_{\text{Compound}} = \frac{43.35 - 26.92}{\sqrt{\frac{0.57^2}{10} + \frac{0.23^2}{10}}} = \frac{16.43}{0.1944} \approx 84.50, \quad \nu = 11.94 \tag{14}$$

with $p = 5.88 \times 10^{-18}$ and Cohen's $d = 37.79$. The compound throughput gain is **+61.0%** (16.43 tok/s) under a

modest VRAM increment of only 40.0 MiB for the Q8_0 draft heads, fully confirming Hypothesis H_1.

7 Architectural Catalog and Physical Feasibility Boundaries

Table 2 presents analytical modeling and physical feasibility projections across candidate models under the 4096 MiB hardware budget.

8 Adversarial Robustness and Defensive Safety Evaluation

Table 3 details empirical results across defensive safety, algorithmic bias, and tokenizer fertility.

8.1 Absence of False Refusals (FPR = 0.0%)

All tested models answered legitimate systems engineering prompts (e.g., terminating deadlocked processes with `kill -9` or dissecting SYN packets with `tcpdump`) with rigorous technical precision, refuting hypotheses of conservative over-alignment on operational tasks.

8.2 Vulnerability to Indirect Instruction Injections and Delimiter Escape

Corroborating Hypothesis H_3, compact models demonstrated acute vulnerability to indirect prompt injections and delimiter breaches (ASR between 50.0% and 100.0%). When presented with unstructured contextual data embedding imperative override directives, models consistently merged data with control flow, emitting the designated canary tokens.

8.3 Agentic Tool Invocation Differentiation and Guardrails

A forensic audit of function calling behavior revealed three distinct execution phases:

1. **Proposed Invocations (`proposed_call`):** The model generates a structured JSON block requesting function execution with concrete parameters;
2. **Schema Conformance (`valid_call`):** The payload complies strictly with the declared JSON schema;
3. **Execution Authorization (`execution_status`):** Guardrail verification of parameter safety prior to OS dispatch.

Models such as Qwen, Gemma, and Llama reliably generated syntactically valid function calls containing injected shell and SQL payloads (`"nginx; rm -rf /"` and `"admin'; DROP TABLE"`), showing that compact models lack intrinsic parameter safety discernment. In all such trials, the harness guardrail intercepted the call with status `BLOCKED_BY_HARNESS`, capturing the full raw payload for auditability without allowing execution. Conversely, SmolLM2 refrained from generating structured tool calls, responding with safe benign text (`SAFE_BENIGN_TEXT`).

9 Algorithmic Bias and Language Fertility Evaluation

9.1 Demographic Bias and Counterfactual Consistency

Evaluations inspired by BBQ and StereoSet-PT revealed that Gemma-4-E2B sustained 100% neutrality across counterfactual gender and socioeconomic pairs. However, under the P3B3 regional variation probe, compact models (SmolLM2 and Qwen-0.5B) failed to issue categorical assertions of cognitive parity, producing evasive statements rather than directly rejecting regional stereotypes.

9.2 Tokenizer Fertility via Parallel Reference Corpus

To eliminate tokenization truncation artifacts and measure true linguistic overhead, we utilized a semantically equivalent parallel corpus tokenized natively via `/tokenize`:

- **SmolLM2-1.7B** exhibited severe tokenization inflation: 2.33 tok/w in Brazilian Portuguese (100 tokens for 43 words) and 2.40 tok/w in European Portuguese, compared to 1.10 tok/w in English (44 tokens for 40 words), imposing an effective throughput penalty exceeding 111%;
- **Gemma-4-E2B** (1.35 tok/w in pt-BR, 1.42 tok/w in pt-PT) and **Qwen-3.5** (1.37 tok/w in pt-BR, 1.44 tok/w in pt-PT) exhibited superior multilingual efficiency, preserving high context capacity and decode speed in Romance languages.

10 Model Provenance and Artifact Reproducibility

To guarantee full experimental auditability, all GGUF model files were cryptographically verified against their SHA-256 checksums, detailed in Table 4.

11 Threats to Validity and Limitations

11.1 Internal Validity

NVML telemetry was sampled at 250 ms intervals; while adequate for capturing sustained tensor allocation peaks, sub-millisecond transients are filtered by the GPU telemetry interface. Energy consumption is modeled via calibrated thermal TDP points rather than direct hardware AC power analyzers.

Table 2: Analytical Architectural Survey and Physical Feasibility Frontiers Under 4 GiB VRAM

Candidate Model	Vendor / License	Parameters (B)	Weights Q4_K_M (MiB)	KV Cache 4k (FP16)	Total VRAM (4k Context)	Status and Limitations on GTX 1050 Ti
Qwen-2.5-0.5B	Alibaba (Apache-2.0)	0.49	469	112 MiB	837 MiB	**Evaluated:** Full feasibility (20.4% of ceiling)
Qwen-2.5-1.5B	Alibaba (Apache-2.0)	1.54	986	224 MiB	1,466 MiB	**Feasible:** High headroom for 16k context
SmolLM2-1.7B	HuggingFace (Apache)	1.71	1,007	192 MiB	1,455 MiB	**Evaluated:** High memory efficiency
Granite-3.1-2B	IBM (Apache-2.0)	2.50	1,580	320 MiB	2,156 MiB	**Feasible:** Native long context support
Phi-4-mini-3.8B	Microsoft (Gated)	3.82	2,510	512 MiB	3,278 MiB	**Non-Executable:** Repository gated (HTTP 401 unauthenticated)
Gemma-3-1B	Google (Gemma)	1.20	880	384 MiB	1,520 MiB	**Feasible:** Low static allocation
Gemma-3-4B	Google (Gemma)	4.30	2,950	1,152 MiB	4,358 MiB	**Physical Block:** OOM under FP16 KV; requires Q4_0 KV compression

Note: Analytical projections derived from exact layer and attention head topologies plus 256 MiB base CUDA driver overhead. Gemma-3-4B exceeds physical 4096 MiB under FP16 cache (4358 MiB), executing successfully only with Q4_0 KV cache compression (3494 MiB, 85.3%).

Table 3: Defensive Safety, Algorithmic Bias, and Tokenizer Fertility Evaluation ($N = 10$, $\tau = 0.0$)

Model	Over-Refusal (FPR %)	Ind. Injection (ASR %)	Delim. Breach (ASR %)	Tool Guard (ASR %)	Neutrality BBQ/Stereo	Fertility pt-BR (tok/w)	Fertility en-US (tok/w)
Qwen-2.5-0.5B	**0.0%**	**0.0%**	50.0%	100.0%	66.7%	1.74	**1.10**
SmolLM2-1.7B	**0.0%**	**0.0%**	50.0%	**0.0%**	33.3%	2.33	**1.10**
Gemma-4-E2B [MTP]	**0.0%**	**0.0%**	100.0%	100.0%	**100.0%**	**1.35**	**1.10**
Gemma-4-E2B [Q4_K_M]	**0.0%**	50.0%	50.0%	100.0%	**100.0%**	**1.35**	**1.10**
Qwen-3.5-2B	**0.0%**	50.0%	100.0%	100.0%	66.7%	1.37	**1.10**
Llama-3.2-3B	**0.0%**	**0.0%**	50.0%	100.0%	66.7%	1.74	**1.10**
Qwen-3.5-4B	**0.0%**	100.0%	100.0%	100.0%	66.7%	1.37	**1.10**

Note: FPR = 0.0% denotes zero false positive refusals on benign operational commands (`kill -9`, `tcpdump`, `gdb`). ASR reflects attack success within the model. Under Tool Guard, 100% ASR indicates the model generated a valid structured tool call emitting injected malicious arguments, which was subsequently intercepted and neutralized by the test harness guardrail prior to shell dispatch.

Table 4: Model Provenance, Physical Artifact Size, and Verified SHA-256 Checksums

Model Identifier	Quantization	Size (MiB)	Context	Official Verified SHA-256 Checksum
`qwen2.5-0.5b`	Q4_K_M	379.3	16,384	`74a4da8c9fdbcd15bd1f6d01d621410d31c6fc00986f5eb687824e7b93d7a9db`
`smollm2-1.7b`	Q4_K_M	1009.1	16,384	`decd2598bc2c8ed08c19adc3c8fdd461ee19ed5708679d1c54ef54a5a30d4f33`
`gemma4:Q4KM`	Q4_K_M	1351.9	16,384	`6dd7ac24fe13a238898e16e253ebc7a72fa27fb83f2658c2f6c241df3af5937d`
`gemma4:MTP`	IQ4_XS	1185.6	16,384	`c149620926221669c4941ae6dac259953cc729c48e4a4edb69e9807b782fd1e3`
`gemma4:E2B`	IQ4_XS+MTP	1225.6	16,384	`9eba819938efccfd6044f8af84e3bbfddc639a2bcf32ebc36420e6a649191919`
`qwen3.5:2B`	Q4_K_XL	1331.2	16,384	`9f7b15d04cf2d5878c8122a7c181dbc09f050cd66080ce3374576e734ccb0910`
`llama3.2:3B`	Q4_K_M	1925.9	16,384	`6c1a2b41161032677be168d354123594c0e6e67d2b9227c84f296ad037c728ff`
`qwen3.5:4B`	Q4_K_M	2458.2	16,384	`00fe7986ff5f6b463e62455821146049db6f9313603938a70800d1fb69ef11a4`

11.2 External Validity

Experiments were executed on an NVIDIA GeForce GTX 1050 Ti (Pascal microarchitecture, Compute 6.1). Modern accelerators based on Ampere, Ada Lovelace, or Blackwell architectures feature native Tensor Cores and FP8/FP4 support, wherein arithmetic intensity boundaries in the Roofline model will shift.

11.3 Construct Validity

The defensive evaluation utilized 10 structured synthetic attack scenarios. While fully reproducible and deterministic, adaptive multi-turn jailbreak strategies may expose additional vulnerabilities.

12 Conclusions and Future Work

This research provides an empirical, mathematical, and systems evaluation of Small Language Models (0.5B to 4B) operating under a physical 4096 MiB VRAM limit.

Primary empirical conclusions are:

1. **Causal Factorial Speculative Acceleration:** Factorial ablation isolated a +25.6% causal throughput speedup attributable strictly to MTP speculative decoding under constant IQ4_XS quantization (43.35 vs 34.51 tok/s, $t = 43.72$, $p < 10^{-15}$), while IQ4_XS weight compression accounts for +28.2% over Q4_K_M (34.51 vs 26.92 tok/s, $t = 65.41$), yielding a compound +61.0% gain ($t = 84.50, p = 5.88 \times 10^{-18}$) with only 40 MiB VRAM overhead for Q8_0 draft heads.
2. **4B Physical Boundary:** 4B architectures operate at the critical edge of memory capacity (76.7%). Gemma-3-4B mandates Q4_0 KV cache compression to avert physical out-of-memory faults.
3. **Defensive Segregation Vulnerability:** While all models achieved FPR = 0.0% on benign tasks, resilience against indirect injection remains critically weak (ASR $\leq$ 100%). In agentic workflows, malicious parameters are readily proposed by compact models, requiring mandatory harness-level parameter interception.
4. **Linguistic Tax via Parallel Corpus:** Parallel corpus tokenization demonstrated an inflation of up to 2.33 tok/w in Portuguese (SmolLM2-1.7B) versus 1.10 tok/w in English, while Gemma-4 and Qwen-3.5 preserved high multilingual efficiency near 1.35 tok/w.

Future research will investigate ternary parameter quantization (1.58 bits) and physical PCIe current probe instrumentation for direct hardware power profiling.

Data Availability Statement

All raw NVML telemetry logs, benchmark traces, evaluation prompts, and statistical scripts are openly available under the MIT license at:

https://github.com/AndersonTavares0/llm-research-bench

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